Sri Sivasubramaniya Nadar College of Engineering, Kalavakkam 603 110

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Department of Computer Science and Engineering

Micro Project

Regulations R2024

Degree and Branch	B.E. Computer Science and Engineering	**Semester**	V
Course Code and Name	**UCS3564 Natural Language Processing**		
Academic Year	2026 2027 ODD	**Batch**	2024 2028
Units Covered	All Units, K3 and K5	**Weightage**	9 percent, of which final demo 5 percent
Team Size	Three students per team	**Maximum**	**100 Marks**

Course Outcomes, PI and WK Mapping

CO	Statement (WK and WP)	K	PO and PSO	Performance Indicator
CO1	Apply text preprocessing techniques and build the language models (WK3, WK4)	K3	PO1, PO5, PSO1	1.4.1, 5.2.2, 13.2.1
CO2	Apply basic levels of knowledge at word level and syntax level in language processing (WK3, WK4)	K3	PO1, PO5, PSO1	1.4.1, 5.2.2, 13.2.1
CO3	Apply computational methods in lexical and vector semantics (WK3, WK4, WK6)	K3	PO1, PO2, PO4, PO5, PO11, PSO1	1.4.1, 2.3.1, 4.3.1, 5.2.2, 12.3.1, 13.4.1
CO4	Explain discourse processing and develop machine translation systems (WK3, WK4, WK6)	K3	PO1, PO2, PO3, PO4, PO5, PSO1, PSO2	1.4.1, 2.2.4, 3.4.2, 4.3.2, 5.2.2, 13.3.2, 14.5.1
CO5	Apply transformers and bidirectional transformer encoders for solving classification and sequence labeling and evaluate on real time applications (WK3, WK4, WK6, WP1, WP2, WP4)	K5	PO1 to PO11, PSO1, PSO2	1.4.1, 2.4.2, 3.4.3, 4.3.4, 5.3.2, 7.1.2, 9.3.1, 10.2.2, 11.3.1, 12.2.1, 13.4.2, 14.4.2

The micro project is the complex engineering problem of this course. It addresses WP1 depth of knowledge, WP2 range of conflicting requirements and WP4 familiarity of issues, along with EA2 level of interaction and EA3 innovation. It maps to SDG 4 Target 4.4 and SDG 9 Target 9.5.

Micro Project Statement

Design and implement a complete Natural Language Processing system for a real time application. The system must run end to end from raw text to a working output, and must include at least one transformer based component. Teams may either define their own problem from the themes below or participate in an open shared task.

Suggested themes

1. Multilingual hate speech or offensive language detection for Tamil and English code mixed text
2. Claim detection and fact verification for news or social media posts
3. Clinical or biomedical named entity recognition and relation extraction
4. Neural machine translation for a low resource Indian language pair, with terminology constraints
5. Question answering over a domain specific document collection using dense retrieval
6. Abstractive summarization of legal judgements or research abstracts
7. Resume to job description matching and skill extraction
8. Conversational assistant for a campus service, with intent detection and slot filling
9. Aspect based sentiment analysis of product or restaurant reviews
10. Grammatical error correction or automated short answer grading

Shared task option: teams may instead register for FIRE 2026 or the WMT 2026 Terminology Task and submit a working system to that task. The task report submitted to the organisers may be used as the project report.

Work Expected

i. Formulate the problem. State the application, the input and output, the users, the real world need and the measurable success criteria. Survey at least five recent papers and state the gap you address. *(CO1 to CO5)*

ii. Prepare the data. Collect or select the corpus, document its licence and size, define the annotation scheme if you label data, and fix the train, validation and test split. *(CO1)*

iii. Design the system. Draw the pipeline architecture with clear modules for preprocessing, representation, modelling, inference and evaluation, and justify every technology choice. *(CO1 to CO5)*

iv. Implement the baseline. Build a classical baseline using TF-IDF or word embeddings with a traditional classifier, or a rule based baseline where classification does not apply. *(CO1, CO3)*

v. Implement the transformer component. Fine tune or apply a pretrained transformer such as BERT, RoBERTa, IndicBERT, mBART or a sequence to sequence model, as suited to the task. Report the model card, hyper parameters and compute used. *(CO5)*

vi. Evaluate and compare. Report accuracy, precision, recall and F1 for classification, or BLEU, chrF and human judgement for translation, or exact match and F1 for question answering. Compare the baseline against the transformer with an error analysis of at least ten failure cases. *(CO4, CO5)*

vii. Handle edge cases and constraints. Demonstrate behaviour on empty input, out of vocabulary and code mixed text, very long input beyond the model window, class imbalance, and unseen classes at inference. State the latency and memory needed for one inference. *(CO5)*

viii. Deliver and present. Wrap the system in a simple interface such as a Streamlit or Gradio application or a REST endpoint, demonstrate it live, and present the work in eight minutes followed by questions. *(CO5)*

Evaluation Criteria

Criterion	Weightage
Problem statement and formulation: clarity, scope and real world relevance	10 percent
Design: architecture and pipeline design, module breakdown, technology choices	25 percent
Implementation: correctness and a functioning end to end pipeline	25 percent
Handling of edge cases and variations	10 percent
Testing: test cases and validation of outputs against expected results	10 percent
Demo and presentation	10 percent
Teamwork and on time submission of proposal, review and final deliverable	10 percent
Total	**100 percent, scaled to 9 marks**

Deliverables

- Proposal of two pages with the problem, dataset, planned architecture, evaluation metrics and work split.
- Project report in IEEE conference format of six to eight pages with the architecture diagram, results tables and error analysis.
- Source code as a GitHub repository with a README, requirements.txt, a reproducible run script and a fixed random seed.
- Working demo application or endpoint, along with a five minute recorded walkthrough.
- Presentation slides and a one page note on the contribution of each team member.

Every team must take a different problem. Pretrained models and public datasets may be used, provided the licence permits it and the source is cited. Code generated by any automated assistant must be declared in the README. Each member must be able to explain the whole system during the viva, not only the part they built.

Prepared By	Verified By	Approved By
(Course In charge)	PAC	Head of the Department